

Multiple Choice (10 marks)

1. Which situation could be represented by the expression 6×2 ?
 - (a) Rocco read six books in two months.
 - (b) Rocco hiked six miles each day for two days.
 - (c) Rocco spent six hours on homework for two days.
 - (d) Rocco has six siblings and two of them are girls.
 - (e) Rocco had a total of six tennis balls in two cans.
2. Find the interval in which the smallest positive root of the following equations lies:
 $x^3 - x - 4 = 0$. Determine the roots correct to two decimal places using the bisection method
 - (a) 2.5
 - (b) 1.7
 - (c) 1.9
 - (d) 2.2
 - (e) 1.8
3. Find the generator for the finite field $\mathbb{Z}_7$.
 - (a) 1
 - (b) 8
 - (c) 0
 - (d) 7
 - (e) 5
4. A company makes 5 blue cars for every 3 white cars it makes. If the company makes 15 white cars in one day, how many blue cars will it make?
 - (a) 25
 - (b) 35
 - (c) 17
 - (d) 13
 - (e) 9
5. A worker on an assembly line takes 7 hours to produce 22 parts. At that rate how many parts can she produce in 35 hours?

- (a) 110 parts
- (b) 245 parts
- (c) 770 parts
- (d) 220 parts
- (e) 140 parts

True / False (10 marks)

Answer True or False

6. True or False: The correct answer to 'Find the sum of $\sum_{n=1}^{\infty} \frac{2}{n^2+4n+3}$ ' is '0.8333'.
7. True or False: In a group, the equation $(ab)^{-2} = b^{-2}a^{-2}$ is false for all groups, and $(ab)^n = a^n b^n$ is also false for all groups.
7. True or False: The estimated sum of 153 and 44 falls within the range of 400 and 599.
8. True or False: The value of the integral $\int_2^4 \frac{\sqrt{\log(9-x)}}{\sqrt{\log(9-x)} + \sqrt{\log(x+3)}} dx$ is 2.2.
9. True or False: The correct answer to 'Compute $\int_{|z|=1} z^2 \sin(1/z) dz$. The answer is Ai with i denoting the imaginary unit, what is A ?' is '-0.5'.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. How many two-digit prime numbers have a units digit of 7?
12. How many paths are there from A to B on the lines of the grid shown, if every step must be up or to the right?

```
[asy]size(3 cm,3 cm);int w=6;int h=3;int i;for (i=0; i<h; ++i){ draw((0,i) - (w-1,i));}for (i=0; i<w; ++i){ draw((i, 0)-(i,h-1));}label("B", (w-1,h-1), NE);label("A", (0,0), SW);[/asy]
```

For example, here is one such valid path highlighted in blue:

```
[asy] size(3 cm,3 cm); int w=6; int h=3; int i; for (i=0; i<h; ++i){ draw((0,i) - (w-1,i)); }for (i=0; i<w; ++i){ draw((i, 0)-(i,h-1)); } label("B", (w-1,h-1), NE); label("A", (0,0), SW); draw((0,0)-(1,0), blue+1.5); draw((1,0)-(1,1), blue+1.5); draw((1,1)-(2,1), blue+1.5); draw((2,1)-(3,1), blue+1.5); draw((3,1)-(4,1), blue+1.5); draw((4,1)-(4,2), blue+1.5); draw((4,2)-(5,2), blue+1.5); [/asy]
```

End of Paper